1874 The year that the **Eads Bridge**, the **world's first bridge** all **made of steel**, opened for traffic across **the Missouri River**.

Alloys

An alloy is a mixture of at least two different elements, at least one of which is a metal. Alloys are used to make many things, including car and airplane parts, musical instruments, jewelry, and medical implants.

In many alloys, all the elements are metal. However, some alloys contain non-metals, such as carbon. The ingredients of an alloy are carefully chosen for the properties they bring to the alloy, whether to make it stronger, more flexible, or rust-resistant. All alloys have metallic properties, are good electrical conductors, and have advantages over pure metals.

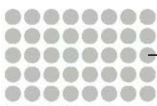
Early alloys

The first man-made alloy was bronze. It was developed around 5,000 years ago by smelting (heating) copper and tin together. This was the start of the Bronze Age, a period in which this new, strong alloy revolutionized the making of tools and weapons. Some thousand years later, people learned to make brass from copper and zinc.

Bronze weapons
Bronze can be hammered thin, stretched, and molded. These objects, made in Mesopotamia around 2000 BCE, were designed to fit on a mace (a clublike weapon).

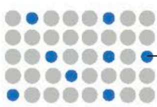
Atomic arrangements

It is how the atoms are arranged in a material that decides how it behaves in different conditions. Atoms of pure metals are regularly arranged, but in alloys this arrangement is disrupted. The atoms of the main component of an alloy may be of a similar size, or much bigger, than those of the added one. They can be arranged in several ways.



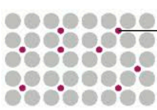
Identical atoms of pure metals

Pure metals
The atoms in a pure metal such as gold (left) are neatly arranged. Under pressure, they will slide over one another, causing cracking.



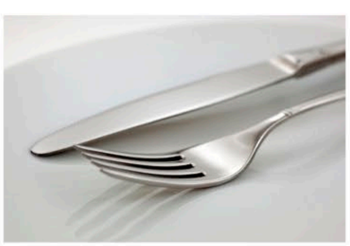
Zinc atoms replace copper atoms in a brass alloy used for trumpets.

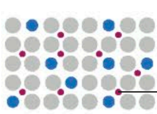
Substitutional alloys
Atoms of the added component take up almost the same space as atoms of the main one. This distorts the structure and makes it stronger.



Tiny carbon atoms sit between large iron atoms, making steel very strong.

Interstitial alloys
These alloys, such as steel used for bridges, are strong: smaller atoms fill the gaps between larger ones, preventing cracking or movement.





Interstitial carbon atoms and substitutional nickel or chromium atoms make stainless steel strong as well as non-rusting.

Combination alloys
Some alloys have a combination of atom arrangements to improve their properties. An example is stainless steel, used in cutlery.

Alloys in coins

Coins used to be made of gold and silver, but these metals are too expensive and not hard-wearing enough for modern use. Several different alloys are used for coins today. They are selected for their cost, hardness, color, density, resistance to corrosion, and for being recyclable.

EU €2-coin
Outer ring: copper (75%), nickel (25%). Center: copper (75%), zinc (20%), nickel (5%).

British £1-coin
Outer ring: copper (76%), zinc (20%), nickel (4%). Inner ring: copper (75%), nickel (25%).

Egyptian £1-coin
Outer ring: steel (94%), copper (2%), nickel plating (4%). Inner ring: steel (94%), nickel (2%), copper plating (4%).

Australian $1-coin
Copper (92%), nickel (2%), aluminum (6%).